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HOSPITAL DATABASE MANAGEMENT SYSTEM

A hospital management system stores information related to patients, doctors, appointments, and billing. The database consists of the following four relations.

RELATION SCHEMA DESCRIPTION

1. Patient

Patient has the following attributes:

- patient_id (INT) – Primary Key
- patient_name (VARCHAR(50))
- age (INT)
- gender (VARCHAR(10))
- city (VARCHAR(50))

2. Doctor

Doctor has the following attributes:

- doctor_id (INT) – Primary Key
- doctor_name (VARCHAR(50))
- specialization (VARCHAR(50))
- experience (INT)

3. Appointment

Appointment has the following attributes:

- appointment_id (INT) – Primary Key
- patient_id (INT) – Foreign Key referencing Patient(patient_id)
- doctor_id (INT) – Foreign Key referencing Doctor(doctor_id)
- appointment_date (DATE)

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4. Bill

Bill has the following attributes:

- bill_id (INT) – Primary Key
- appointment_id (INT) – Foreign Key referencing Appointment(appointment_id)
- amount (INT)
- payment_status (VARCHAR(20))

TASK–1: TABLE CREATION

Write SQL statements to create the above tables with appropriate primary key and foreign key constraints.

TASK–2: DATA INSERTION

Insert the following data into the respective tables.

```
INSERT INTO Appointment VALUES
(1001, 1, 101, '2024-02-10'),
(1002, 2, 105, '2024-02-11'),
(1003, 3, 102, '2024-02-12'),
(1004, 4, 103, '2024-02-13'),
(1005, 5, 101, '2024-02-14');
```

```
INSERT INTO Bill VALUES
(5001, 1001, 2000, 'Paid'),
(5002, 1002, 1500, 'Pending'),
(5003, 1003, 3000, 'Paid'),
(5004, 1004, 4000, 'Paid'),
(5005, 1005, 2500, 'Pending');
```

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INSERT INTO Patient VALUES
(1, 'Rahul Sharma', 35, 'Male', 'Delhi'),
(2, 'Anita Verma', 28, 'Female', 'Mumbai'),
(3, 'Suresh Kumar', 42, 'Male', 'Chennai'),
(4, 'Priya Singh', 30, 'Female', 'Kolkata'),
(5, 'Amit Das', 50, 'Male', 'Bhubaneswar');

INSERT INTO Doctor VALUES
(101, 'Dr. Mehta', 'Cardiology', 15),
(102, 'Dr. Rao', 'Orthopedics', 10),
(103, 'Dr. Sen', 'Neurology', 12),
(104, 'Dr. Gupta', 'Dermatology', 8),
(105, 'Dr. Iyer', 'General Medicine', 20);

TASK–3: SQL QUERIES (USING JOINS)

Write SQL queries to perform the following:

1. Display patient name, doctor name, and appointment date for all appointments.
2. List the names of patients who consulted a doctor from the Cardiology specialization.
3. Display patient name, doctor name, and bill amount for all paid bills.
4. Find the total bill amount collected by each doctor.
5. List the names of patients whose bill payment status is pending.
6. Display doctors who have treated patients aged more than 40 years.
7. Count the number of appointments handled by each doctor.

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